EXPERIMENT 07

Part-of-Speech Tagging using NLTK Library

AIM

To tokenize an English sentence and assign grammatical Part-of-Speech (POS) tags to each token using the NLTK library.

PROCEDURE

1. Import `nltk`, `word_tokenize`, and `pos_tag`.
2. Prepare a diverse test sentence containing verbs, nouns, auxiliaries, adjectives, and prepositions.
3. Tokenize the sentence into individual word and punctuation tokens.
4. Invoke `nltk.pos_tag()` to label tokens with Penn Treebank POS identifiers.
5. Display each token alongside its grammatical tag and standard English descriptor.

PROGRAM

```
import nltk

nltk.download('punkt', quiet=True)
nltk.download('averaged_perceptron_tagger', quiet=True)
nltk.download('tagsets', quiet=True)

sentence = "The intelligent system quickly parsed the complex linguistic queries."

tokens = nltk.word_tokenize(sentence)
tagged_tokens = nltk.pos_tag(tokens)

tag_explanations = {
    'DT': 'Determiner',
    'JJ': 'Adjective',
    'NN': 'Noun, singular',
    'RB': 'Adverb',
    'VBD': 'Verb, past tense',
    'NNS': 'Noun, plural',
    '.': 'Punctuation mark'
}

print(f"{'Token':<14} | {'POS Tag':<8} | {'Description'}")
print("-" * 45)
for word, tag in tagged_tokens:
    desc = tag_explanations.get(tag, 'Lexical Category')
    print(f"{word:<14} | {tag:<8} | {desc}")
```

OUTPUT

```
> python3 experiment_07.py
```

```
Token          | POS Tag  | Description
-------------------------------------------
The            | DT       | Determiner
intelligent    | JJ       | Adjective
system         | NN       | Noun, singular
quickly        | RB       | Adverb
parsed         | VBD      | Verb, past tense
the            | DT       | Determiner
complex        | JJ       | Adjective
linguistic     | JJ       | Adjective
queries        | NNS      | Noun, plural
.              | .        | Punctuation mark
[Process completed with exit code 0]
```

RESULT

Thus, Part-of-Speech tagging was successfully executed on natural language text using the pre-trained NLTK POS tagger.